ELITE IGCSE MATHEMATICS
EXPERIENCE NOTES

Combined & Conditional Probability

Learn the trigger, write the first line, then solve one similar problem while the method is still fresh.

COMBINED & CONDITIONAL PROBABILITY · CH6 · L16

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The Map

This lesson collapses Combined & Conditional Probability into the few exam moves that keep repeating. Read the trigger, copy the first line, then practise the same idea on the next page.

01 Count wanted over total

TRIGGER *"probability", "random", "given that"*

02 Use complement

TRIGGER *"probability", "random", "given that"*

03 Multiply along tree branches

TRIGGER *"probability", "random", "given that"*

04 Use conditional probability

TRIGGER *"probability", "random", "given that"*

05 Calculate probability

TRIGGER *"probability", "random", "given that"*

06 Use the complement

TRIGGER *"probability", "random", "given that"*

BANK EVIDENCE 11 website questions, 11 checked solutions, 3 local PDFs read.

01 Count wanted over total

TRIGGER *"probability", "random", "given that"*

BECOMES A Combined & Conditional Probability question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$P(\text{all 3 red are taken}) = \frac{3}{12} \cdot \frac{2}{11} \cdot \frac{1}{10}$$

SIMPLEST STRATEGY

1. Underline the target: what is the question asking you to find?
2. Write the rule, theorem, or equation before any calculator work.
3. Substitute one value at a time and keep units/unknowns visible.
4. Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

> **Source pattern:** Jun 2023 P2H Q20
>
> **There are 3 red counters**
> This is a reasoning step: write one clear sentence, then continue the calculation.
>
> **At least one red counter**
> This is a reasoning step: write one clear sentence, then continue the calculation.
>
> **Finish:** $\frac{219}{220}$

FORMULA BANK

Conditional probability: $P(A|B) = \frac{P(A \cap B)}{P(B)}$. Without replacement changes the denominator.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 01

SAME IDEA Use the same first line from Strategy 01.

QUESTION

> A bag has 5 red counters and 7 blue counters. Two counters are taken without replacement. Find the probability that both are red.

YOUR SOLUTION

02 Use complement

TRIGGER *"probability", "random", "given that"*

BECOMES A Combined & Conditional Probability question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$P(\text{odd}) = \frac{1}{2}$$

SIMPLEST STRATEGY

1. Underline the target: what is the question asking you to find?
2. Write the rule, theorem, or equation before any calculator work.
3. Substitute one value at a time and keep units/unknowns visible.
4. Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

> **Source pattern:** Jun 2022 P2HR Q14
>
> **The complement of at least**
>
> $$P(\text{odd}) = \frac{1}{2}$$
>
> **Finish:** $\frac{3}{4}$.

FORMULA BANK

Conditional probability: $P(A|B) = \frac{P(A \cap B)}{P(B)}$. Without replacement changes the denominator.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 02

SAME IDEA Use the same first line from Strategy 02.

QUESTION

> A bag has 5 red counters and 7 blue counters. Two counters are taken without replacement. Find the probability that both are red.

YOUR SOLUTION

03 Multiply along tree branches

TRIGGER *"probability", "random", "given that"*

BECOMES A Combined & Conditional Probability question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$P(\text{all 3 red are taken}) = \frac{3}{12} \cdot \frac{2}{11} \cdot \frac{1}{10}$$

SIMPLEST STRATEGY

1. Underline the target: what is the question asking you to find?
2. Write the rule, theorem, or equation before any calculator work.
3. Substitute one value at a time and keep units/unknowns visible.
4. Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

> **Source pattern:** Jun 2023 P2H Q20
>
> **There are 3 red counters**
> This is a reasoning step: write one clear sentence, then continue the calculation.
>
> **At least one red counter**
> This is a reasoning step: write one clear sentence, then continue the calculation.
>
> **Finish:** $\frac{219}{220}$

FORMULA BANK

Conditional probability: $P(A|B) = \frac{P(A\cap B)}{P(B)}$. Without replacement changes the denominator.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 03

SAME IDEA Use the same first line from Strategy 03.

QUESTION

> A bag has 5 red counters and 7 blue counters. Two counters are taken without replacement. Find the probability that both are red.

YOUR SOLUTION

04 Use conditional probability

TRIGGER *"probability", "random", "given that"*

BECOMES A Combined & Conditional Probability question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$P(\text{win tennis}) = 0.6$$

SIMPLEST STRATEGY

1. Underline the target: what is the question asking you to find?
2. Write the rule, theorem, or equation before any calculator work.
3. Substitute one value at a time and keep units/unknowns visible.
4. Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1HR Q16

Calculate probability

$$P(\text{win tennis}) = 0.6$$

Calculate probability

$$0.42 \div 0.6 = 0.7$$

Finish: 0.12.

FORMULA BANK

Conditional probability: $P(A|B) = \frac{P(A \cap B)}{P(B)}$. Without replacement changes the denominator.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 04

SAME IDEA Use the same first line from Strategy 04.

QUESTION

> A bag has 5 red counters and 7 blue counters. Two counters are taken without replacement. Find the probability that both are red.

YOUR SOLUTION

05 Calculate probability

TRIGGER *"probability", "random", "given that"*

BECOMES A Combined & Conditional Probability question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$P(\text{win tennis}) = 0.6$$

SIMPLEST STRATEGY

1 Underline the target: what is the question asking you to find?

2 Write the rule, theorem, or equation before any calculator work.

3 Substitute one value at a time and keep units/unknowns visible.

4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1HR Q16

Calculate probability

$$P(\text{win tennis}) = 0.6$$

Calculate probability

$$0.42 \div 0.6 = 0.7$$

Finish: 0.12.

FORMULA BANK

Conditional probability: $P(A|B) = \frac{P(A \cap B)}{P(B)}$. Without replacement changes the denominator.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 05

SAME IDEA Use the same first line from Strategy 05.

QUESTION

> A bag has 5 red counters and 7 blue counters. Two counters are taken without replacement. Find the probability that both are red.

YOUR SOLUTION

06 Use the complement

TRIGGER *"probability", "random", "given that"*

BECOMES A Combined & Conditional Probability question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$P(\text{all 3 red are taken}) = \frac{3}{12} \cdot \frac{2}{11} \cdot \frac{1}{10}$$

SIMPLEST STRATEGY

1. Underline the target: what is the question asking you to find?
2. Write the rule, theorem, or equation before any calculator work.
3. Substitute one value at a time and keep units/unknowns visible.
4. Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

> **Source pattern:** Jun 2023 P2H Q20
>
> **There are 3 red counters**
> This is a reasoning step: write one clear sentence, then continue the calculation.
>
> **At least one red counter**
> This is a reasoning step: write one clear sentence, then continue the calculation.
>
> **Finish:** $\frac{219}{220}$

FORMULA BANK

Conditional probability: $P(A|B) = \frac{P(A \cap B)}{P(B)}$. Without replacement changes the denominator.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 06

SAME IDEA Use the same first line from Strategy 06.

QUESTION

> A bag has 5 red counters and 7 blue counters. Two counters are taken without replacement. Find the probability that both are red.

YOUR SOLUTION

Quick Reference

TRIGGER → FIRST MOVE

WHEN YOU SEE	WRITE THIS FIRST
"probability", "random", "given that"	Count wanted over total
"probability", "random", "given that"	Use complement
"probability", "random", "given that"	Multiply along tree branches
"probability", "random", "given that"	Use conditional probability
"probability", "random", "given that"	Calculate probability
"probability", "random", "given that"	Use the complement

FINAL BOTTOM LINE Do not try to remember every past-paper wording. Remember the trigger, write the first line, and let the working follow.